COSAT 2.0 Cosmology

Abstract

COSAT 2.0 (Complex Observable Spacetime Asymmetry Theory) is a mechanically grounded unified field theory that resolves the incompatibilities between Quantum Mechanics and General Relativity. By substituting zero-dimensional point particles and virtual exchange bosons with 1D topological tethers interacting within a **6D *SO(3,3)*** fluid metric, COSAT unifies all four fundamental forces as geometric variations of a single electro-gravitational phase state. Furthermore, COSAT replaces dark matter, dark energy, and mathematical singularities with rigorous hydrostatic fluid dynamics, projective geometry, and a thermodynamically driven cosmic meta-cycle.

I. The Mathematical Foundation

Interactions are dictated by the physical intersection of expanding complex phase-waves. The foundational interaction is defined by the Unified Topological Force Polynomial:

$$\mathcal{F}_{knot} = \frac{1}{r^2}\left(q^2 - Gm^2 + 2iqm\sqrt{G}\right)$$

- **Electromagnetic Flavor $(\boldsymbol{q^2/r^2})$:** The baseline topological surface interaction.
- **Gravitational Flavor $(\boldsymbol{-Gm^2/r^2})$**: The residual macroscopic van der Waals electric force.
- **Weak Flavor $\left(\boldsymbol{2iqm\sqrt{G}/r^2}\right)$**: The inertial drag (Transverse Phase-Shear) of the 6D fluid.
- **Strong Flavor:** The maximum structural tension limit of the 1D tethers and internal quark plasma resonance. (Dimensionality is expressed in natural units **(c = G = 1).)**

II. Sub-Atomic & Quantum Phenomena

- **Mass & The Metric Halo:** Mass is an emergent property. The topological knot of a charge polarizes the surrounding 6D metric, trapping a localized solvent cage known as a "Metric Halo."
- **Inertia (The Wave-Center Feedback Loop):** A particle at rest or constant velocity maintains a perfectly symmetric convergent wave boundary. When an external force accelerates the particle, the topological knot is physically displaced from this symmetric center. The 6D metric fluid violently attempts to push the knot back into the dead-center of the wave geometry to restore topological equilibrium. This wave-centering restoring force *is* Inertia.
- **Wave-Particle Duality (Dual-Cone):** A particle exists in two orthogonal cones. The Real Cone (Particle) is localized upon measurement. The Imaginary Cone (Wave) superluminally projects continuous phase-waves.
- **Entanglement (Phase-Lock):** Distance in the Real Cone is an illusion. Entangled particles are geometrically tethered to the exact same Primary Imaginary Momentum Vector in the 6D bulk. When one collapses, the shared topological anchor dictates the other instantly.

III. Macroscopic & Relativistic Phenomena

- **Gravity (Residual Electric van der Waals):** Gravity is not a fundamental pull. Because of the Pauli exclusion principle, positive and negative charges cannot perfectly superimpose. Treated as macroscopic H_2 super-molecules, celestial bodies geometrically stretch their electron clouds (Feynman-Hellmann distortion) into a static, spin-paired Singlet state, generating a permanent, non-fluctuating residual attractive force.
- **The Speed of Light (*c*):** Not a speed limit, but the maximum phase-velocity the 6D fluid can sustain before topological fracture. Photons exist precisely at the (0,0) Luminal Origin State.
- **Black Holes (Quark Plasma Core):** Mathematical singularities are impossible because perfect superposition yields zero residual gravity. A black hole core is an ultra-dense, finite Quark Plasma.
- **Event Horizons (SCB Evanescence):** The Schwarzschild Boundary is a 2D Real discontinuity. "Macro-Real" information (identity) is annihilated and painted onto the 2D surface. The "Micro-Real" stripped filaments drop into the core, rotating (90°) into an inverted negative metric space.

IV. Cosmological Phenomena

- **The Asymmetry Force (Replacing the Dark Sector):** * *Dark Matter Replaced:* Galaxies are not held together by invisible mass; they are hydrostatically crushed together by the ambient 6D metric pressure (Topological Shielding / Kinetic Shadow).
- ***Dark Energy Replaced:*** Cosmic acceleration is driven by Thermodynamic Drift—matter siphoning violently down the entropy gradient into older, less dense regions of the metric.
- **Quasar Nucleation (Seeding the Universe):** When an accretion disc breaches the SCB at the equator, black holes vent their ultra-dense plasma as relativistic polar jets. These jets rapidly re-hadronize into heavy elements, seeding primordial gas clouds to trigger the ignition of the first stars. Black holes precede stars.
- **Systolic Pre-Seeding (Solving JWST Anomalies):** As a universe collapses, gorging black holes carve deep pressure channels (filaments) into the metric. When the universe expands (Big Bounce), new matter instantly snaps into these pre-existing grooves, forming massive galaxies almost immediately.
- **The Cosmic Meta-Cycle:** With each iteration of the universe, the "Cosmic Web" becomes denser and more complex. Eventually, hyper-accretion causes black holes to devour the universe too quickly, starving the quasars. Without jets to carve new grooves, the black holes merge, sandblasting the metric clean. The next Big Bounce expands into a pristine, smooth metric, starting Iteration 0 over again.

  A Note on Dimensionality: In standard SI units, adding charge (q) to mass (m) is dimensionally impossible, as they are treated as fundamentally different properties. However, advanced theoretical physics frequently utilizes Natural Units (c = G = 1) to bridge this gap. In Phase Kinetics, this summation is natively legal without hidden constants. Because both charge and mass are defined strictly as manifestations of the exact same SO(3,3) fluid momentum flux (polarized pressure vs. unpolarized leakage), they share the exact same foundational geometric dimensions.